VAN LANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY

TRAN NGUYEN TRUNG HIEU – 207CT55052

PROJECT REPORT

ANALYSIS OF LUNG DISEASE THROUGH X-RAY IMAGES USING VISION TRANFORMER

UNDERGRADUATE THESIS REPORT

Major: Information Technology (7480201)

Ho Chi Minh City – 2025

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Advisor: MSc. Phan Ho Viet Truong

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ABSTRACT

Abstract: The rapid advancement of artificial intelligence (AI) has significantly enhanced medical diagnostics, particularly in lung disease detection through chest X-ray analysis. Traditional diagnostic methods often rely on radiologists' expertise, which can be time-consuming and subject to variability. AI-powered solutions, especially deep learning models, have demonstrated remarkable potential in automating and improving diagnostic accuracy. Among various deep learning architectures, the Vision Transformer (ViT) has emerged as a promising alternative to conventional convolutional neural networks (CNNs), such as ResNet50.

This study explores the application of Vision Transformer architectures in lung disease classification using chest X-ray images. Unlike CNNs, ViT utilizes self-attention mechanisms to effectively capture long-range dependencies within image data, enhancing diagnostic precision. In this research, several ViT variants were systematically evaluated, including the traditional ViT architecture and more advanced transformer-based architectures such as Swin Transformer, which incorporates local self-attention and hierarchical processing capabilities suitable for medical imaging. The study's objective was to identify the most effective transformer-based model for lung disease detection, evaluating accuracy, precision, recall, F1-score, and AUC-ROC.

Experimental results demonstrated the superior performance of transformer-based architectures, notably achieving over **90% accuracy and an AUC-ROC above 0.98**. Additionally, Grad-CAM visualization was integrated, providing interpretable insights by highlighting suspected disease regions clearly within the X-ray images.
The best-performing transformer-based model was integrated into a web application, allowing users to upload X-ray images for automated analysis. The system also includes an AI-powered chatbot, offering structured medical feedback based on diagnostic outcomes.

This research contributes significantly to AI-driven medical diagnostics, demonstrating the practical applicability and effectiveness of Vision Transformer architectures in analyzing lung diseases from chest X-rays.

Keywords: AI, Vision Transformer (ViT), Lung Disease Diagnosis, X-ray Analysis, Swin Transformer, Grad-CAM.

DECLARATION

I, **Tran Nguyen Trung Hieu,** hereby declare that all material in this report, entitled **ANALYSIS OF LUNG DISEASE THROUGH X-RAY IMAGES USING VISION TRANSFORMER**, submitted in partial fulfillment of the requirements of the Faculty of Information Technology, School of Technology, Van Lang University, is the result of my work and does not involve plagiarism.

I confirm that any ideas taken from printed sources (books, magazines) or electronic sources (websites) have been properly credited to the original authors. I understand that failure to do so may result in rejection or dismissal.

Ho Chi Minh City, Date: 22/03/2025

Advisor	**Student**
(Signature & Full Name)	*(Signature & Full Name)*

Tran Nguyen Trung Hieu

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Finally, I would like to acknowledge the research community and the developers of open-source tools that have contributed to the advancement of artificial intelligence and medical imaging, which serve as the foundation for this work.

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PHIẾU NHẬN XÉT CỦA GIẢNG VIÊN HƯỚNG DẪN
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Tên đề tài nghiên cứu (Project Title): Phân tích bệnh phổi qua hình ảnh X-quang bằng mô hình Vision Transformer (ANALYSIS OF LUNG DISEASE THROUGH X-RAY IMAGES USING VISION TRANSFORMER) .....

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EVALUATION AND CONCLUSION

#	Tiêu chí đánh giá (Evaluation Criteria)	Nhận xét (Comments)	Đánh giá (Đạt/Không đạt) Assessment (Pass/Fail)
A	**Nhận xét đánh giá về quá trình viết khoá luận tốt nghiệp / Evaluation of the thesis writing process**		
1	*Tinh thần thái độ / Attitude and work ethic*		
2	*Thực hiện kế hoạch làm việc do GVHD quy định / Adherence to the work plan set by the advisor*		
3	*Nộp báo cáo khoá luận tốt nghiệp cho GVHD/Khoa / Submission of the thesis to the advisor/faculty*		
B	**Nhận xét đánh giá về khoá luận tốt nghiệp / Evaluation of the thesis content**		
1	*Hình thức đồ án đã thực hiện / Report format and presentation*		
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Kết luận của GVHD *(Cho phép/Không cho phép sinh viên nộp và bảo vệ khoá luận tốt nghiệp)* **Advisor's Conclusion** *(Approve/Disapprove for submission and defense of the thesis)*	

Ho Chi Minh City, Date: ……/……/2025

Giảng viên hướng dẫn / Advisor

(Ký và ghi rõ họ và tên / Signature & Full Name)

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LIST OF ABBREVIATIONS

No.	Abbreviation	Meaning
1	ViT	Vísion Tranformer
2	AI	Artificial Intelligence
3	CNN	Convolutional Neural Network
4	AUC-ROC	Area Under the Receiver Operating Curve
5	API	Application Programming Interface
6	UI	User Interface
7	F1-score	Harmonic Mean of Precision and Recall

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CHAPTER 1: INTRODUCTION

1.1 Research Rationale

Research Rationale

Lung diseases—particularly pneumonia—remain among the leading causes of global morbidity and mortality. According to the World Health Organization (WHO), pneumonia claims more than 2.5 million lives annually, especially affecting children under five and the elderly. Chest X-ray imaging plays a critical role in early diagnosis, but manual interpretation by radiologists can be time-consuming and prone to variability.

The rise of artificial intelligence (AI), particularly deep learning, has introduced promising opportunities for automating medical image analysis. Among modern architectures, transformer-based models such as the Vision Transformer (ViT) and its hierarchical variant, the Swin Transformer, have shown notable potential in capturing both global and local patterns in images.

This project explores the feasibility of applying transformer-based models—specifically the Swin Transformer—for classifying chest X-ray images into three categories: Normal, Bacterial Pneumonia, and Viral Pneumonia. Beyond model development, the research aims to implement a web-based diagnostic system with an integrated AI chatbot, providing accessible and real-time medical feedback, especially in low-resource settings.

Research Questions

1. How do Vision Transformer variants perform compared to traditional CNN models in classifying lung diseases from chest X-rays?
2. What are the strengths and limitations of deploying transformer-based models on moderately sized medical datasets?
3. How can a transformer-based diagnostic system be integrated into a web platform with a chatbot to support user interaction and deliver real-time medical guidance?

Research Hypotheses

1. Vision Transformer models—especially Swin Transformer—can outperform traditional CNNs like ResNet50 in both accuracy and generalization.
2. Integrating a diagnostic model with an AI chatbot will enhance user experience and improve the accessibility of preliminary lung disease screening.

1.2 Research Objectives

Urgency of the Problem

- The need for accurate, automated lung disease detection to reduce reliance on limited radiologist resources.
- The importance of model interpretability (e.g., heatmap visualizations) to increase clinical trust.
- The necessity of building a real-time, user-friendly system suitable for non-experts and low-resource settings.

General Objective

To develop and evaluate a transformer-based system for automated classification of lung diseases from chest X-ray images, and deploy it as a functional web application with integrated chatbot support for interpretability and user engagement.

Specific Objectives

- Design and train transformer-based models to classify chest X-rays into Normal, Bacterial Pneumonia, and Viral Pneumonia categories
- Compare the performance of the proposed models against a conventional CNN baseline using standard classification metrics.
- Enhance model transparency through visual explanation techniques such as Grad-CAM.
- Implement a complete web-based diagnostic system, including frontend and backend modules, for real-time prediction and visualization.
- Integrate an AI-powered chatbot to deliver context-aware medical suggestions based on model predictions.

1.3 Research Content and System Design

Research Subject

This study focuses on the application of AI-based methods—particularly Vision Transformers—for the automatic classification of lung diseases from chest X-ray images.

Research Scope

- **Data domain:** Frontal chest X-ray images (raw, without automatic lung-region cropping).
- **Model comparison:** Transformer-based architectures (ViT, Swin-Base) versus CNN baseline (ResNet50).
- **System functionality:**
    - Automated classification into Normal, Bacterial Pneumonia, or Viral Pneumonia.
    - Grad-CAM heatmap overlays for interpretability.
    - Real-time web interface for image upload and result display.
    - AI chatbot integration for medical advice.
- **Limitations:**
    - No automated lung-region cropping (images are full-field X-rays).
    - No clinical trial validation during this project—data derived from publicly available datasets.
    - Database storage of user data and historical results is not implemented.

Research Content

The research is structured around the following core components:

- **Data Processing**: Publicly available datasets are cleaned, standardized, and augmented to improve generalization during model training.
- **Model Development and Evaluation**: Transformer-based architectures are trained and compared with a CNN baseline using standard classification metrics.
- **Model Interpretability**: Grad-CAM visualizations are incorporated to highlight important regions influencing the model's predictions.
- **System Implementation**: A web application is developed that supports image upload, displays predictions and heatmaps, and features an AI-driven chatbot for user guidance.

1.4 Research Method

Theoretical Research Methods

- **Literature Review:** Examine existing AI applications in medical imaging, focusing on CNNs (e.g., ResNet, DenseNet, CheXNet) and transformer architectures (ViT, Swin Transformer).
- **Model Architecture Analysis:** Study self-attention, residual and dense connections, hierarchical feature extraction, and interpretability techniques (e.g., Grad-CAM).
- **Hypothesis Formulation and Evaluation:** Develop testable hypotheses regarding model performance and integration efficacy; select appropriate metrics and statistical methods (classification metrics, ROC analysis).

Practical Research Methods

- **Dataset Collection and Preprocessing:** Source chest X-ray datasets (e.g., Kaggle Pneumonia dataset), clean data, and apply augmentation.
- **Model Training and Validation:** Fine-tune ResNet50 and Vision Transformer variants on GPU-enabled environments (Google Colab Pro with NVIDIA Tesla T4).
- **System Implementation:**
    - Build and deploy a RESTful API (FastAPI) serving model inference and Grad-CAM.
    - Develop an Angular frontend for user interaction and visualization.
    - Integrate ChatGPT through API calls for automated medical advice.
- **Performance Evaluation:** Generate classification reports, confusion matrices, and ROC curves; conduct comparative analysis.
- **Expert Consultation:** Gather feedback from radiologists and AI researchers to validate interpretability and usability.

1.5 Technical Implementation Overview

- This project extends existing work on CNN-based chest X-ray diagnosis (e.g., CheXNet) by exploring newer Vision Transformer architectures, particularly Swin Transformer, which addresses ViT's limitations (large data requirements, weak local context) through window-based self-attention and hierarchical design.
- Technological solutions applied include:
    - **PyTorch and FastAI** for model development and training.
    - **Timm library** for prebuilt transformer models (DeiT, Swin).
    - **Grad-CAM** for visual interpretability.
    - **Angular** for frontend UI and **FastAPI** for backend inference services.

    - **ChatGPT API** for AI-powered medical chatbot functionality.
- Additional issues addressed:
    - Demonstrating Grad-CAM consistency across development (Google Colab) and deployment (Linux server with FastAPI).
    - Managing GPU memory constraints when fine-tuning Swin-Base.
    - Ensuring a seamless user experience with minimal latency in the web application and chatbot responses.

1.6 Implementation Timeline

Week	Task
1-2	In-depth literature review; finalize dataset sources; plan preprocessing pipeline.
3-4	Data preprocessing and augmentation implementation; exploratory data analysis; baseline ResNet50.
5-6	Fine-tune ResNet50 and evaluate; begin adapting ViT (DeiT) and custom ViT architectures.
7-8	Implement and train Swin-Base model; integrate Grad-CAM visualization; conduct performance comparison.
9-10	Develop FastAPI backend for inference and Grad-CAM; build Angular frontend; integrate APIs.
11	Integrate ChatGPT chatbot; conduct end-to-end tests; collect expert feedback.
12	Final evaluation, write-up of results, prepare for defense; revise based on feedback.

Table 1: Implementation Timeline for the Project

1.7 Expected Outcomes

1.7.1 Trained Transformer-Based Classification Model

At least one Swin Transformer variant achieving ≥90% accuracy and AUC ≥0.95 for three-class classification (Normal, Bacterial, Viral).

1.7.2 Comprehensive Performance Comparison

A detailed comparison of ResNet50, ViT variants, and Swin-Base on metrics: Accuracy, Precision, Recall, F1-score, and AUC-ROC; demonstration of Swin's advantages.

1.7.3 Grad-CAM Visualizations

Interpretable heatmaps for each predicted class that highlight suspicious lung regions; analysis of consistency between development and deployment environments.

1.7.4 Functional Web Application

- Angular-based UI allowing image upload, result display (label, confidence, heatmap), and embedded chatbot.
- FastAPI endpoints serving model inference, Grad-CAM generation, and chatbot queries.

1.7.5 Integrated Medical Chatbot

Context-aware advice based on model output: reassurance for "Normal," referral suggestions for pneumonia cases.

1.7.6 Documentation and Code Repository

- Installation Guide for environment setup and model deployment.
- User Guide explaining usage of the web app, interpretation of results, and chatbot interaction.
- Public GitHub repository containing all source code, trained model weights, and preprocessing scripts.

1.7.7 Foundation for Future Clinical Validation

Recommendations for adding automatic lung-region cropping, optimizing for mobile deployment (ONNX, quantization), and conducting clinical studies with hospital data.

1.8 Thesis Organization

The undergraduate thesis is structured to guide the reader from foundational concepts to system implementation and evaluation:

- **Chapter 1: Introduction**: Introduces background, rationale, research questions, hypotheses, objectives, scope, methods, scientific basis, timeline, expected outcomes, and thesis organization.
- **Chapter 2: Theoretical Framework**: Presents an in-depth literature review of AI in medical imaging (CNNs, DenseNet, CheXNet, Vision Transformer, Swin), describes research methods, and analyzes existing work to identify the research gap.
- **Chapter 3: Current State of the Research Topic**: Explains basic theoretical concepts (detailed model architectures and interpretability techniques) and provides data presentation (dataset description, distribution, limitations) relevant to chest X-ray classification.
- **Chapter 4: Research Results**: Defines system objectives and scope, conducts system analysis (functional hierarchy, data flow), details implementation (developed functions, UI, API integration), presents classification performance (ResNet vs. ViT vs. Swin), shows Grad-CAM visualizations, and discusses theory versus practice and encountered challenges.
- **Chapter 5: Conclusion and Recommendations**: Summarizes research findings (model performance, system functionality, interpretability), highlights contributions, acknowledges limitations (data, deployment constraints), and recommends future directions (lung cropping, multi-disease classification, mobile optimization, clinical validation).
- **References**: Lists all cited literature, including seminal papers on ResNet, Vision Transformer, CheXNet, and recent studies on Transformer applications in medical imaging.
- **Appendices**
    1. **Installation Guide:** Detailed steps for setting up the development environment, installing dependencies, loading the trained model, and deploying the web application.
    2. **User Guide:** Instructions on uploading chest X-ray images, interpreting classification results and Grad-CAM heatmaps, and interacting with the chatbot.
    3. **Source Code Repository:** Link to the public repository containing all code, model files, and preprocessing scripts necessary to reproduce the study's results.

CHAPTER 2: LITERATURE REVIEW

This chapter presents an overview of recent advancements in applying artificial intelligence (AI), particularly deep learning models, to medical image analysis, focusing on lung disease diagnosis using chest X-rays.

2.1 Introduction to the Research Field

Artificial intelligence (AI) has transformed medical imaging by enabling automated, accurate interpretation of radiographs. Early deep learning approaches relied on convolutional neural networks (CNNs) to extract hierarchical features from images:

- **CheXNet (DenseNet-121)**: Fine-tuned on over 100 000 frontal chest X-rays (ChestX-ray14), CheXNet achieved radiologist-level pneumonia detection by using dense connectivity to improve gradient flow and feature reuse. It employed Grad-CAM to generate heatmaps that highlight regions contributing to its predictions.
- **ResNet (Residual Network)**: Introduced skip connections to mitigate vanishing gradients, allowing very deep architectures (e.g., ResNet50) to excel in medical image classification with transfer learning from ImageNet. ResNet50 commonly serves as a strong baseline for chest X-ray diagnostics.

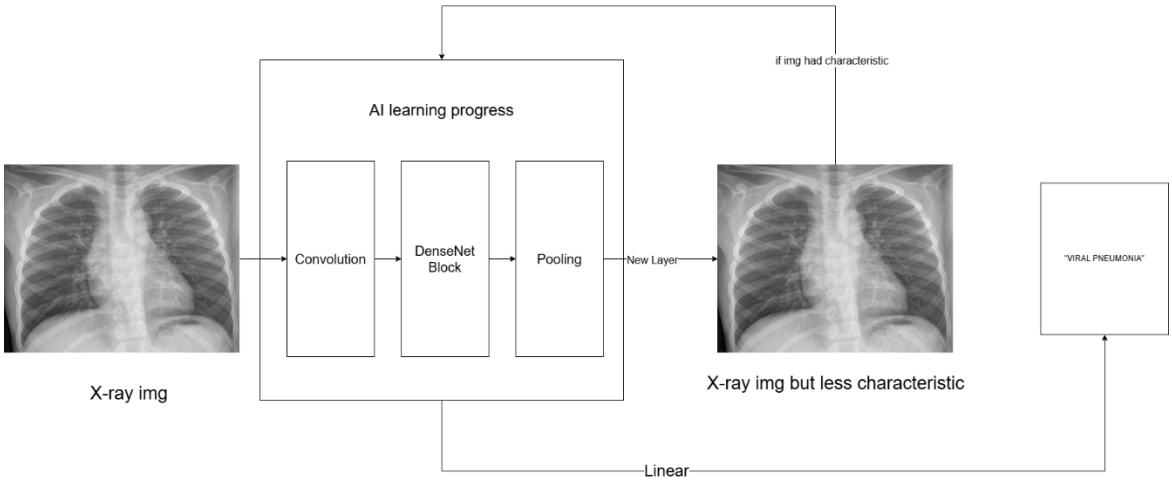

Figure 1 DenseNet-121 Architecture

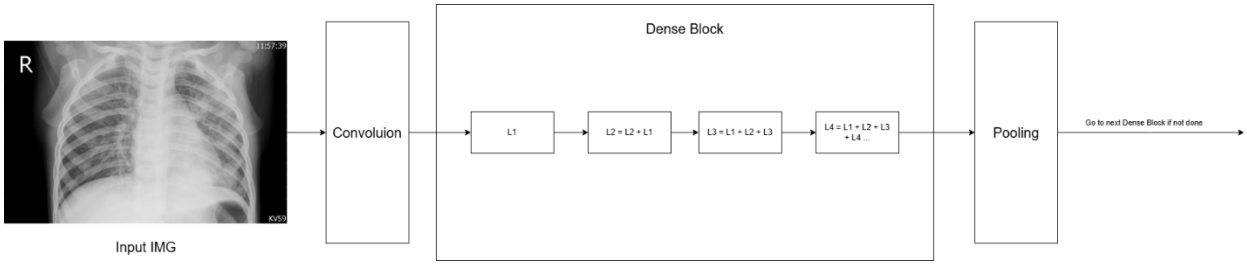

Figure 2 DenseNet Block in DenseNet Architecture

More recently, **Vision Transformer (ViT)** architectures—adapted from NLP transformers—divide an image into patches and process them through self-attention layers, capturing global relationships from the first layer. While vanilla ViT requires large-scale data, its distilled and hierarchical variants have reduced that dependency:

- **DeiT (Distilled ViT)**: Pretrained with knowledge distillation, it performs robustly on medium-sized datasets and can be fine-tuned for binary X-ray classification (e.g., Normal vs. Pneumonia).
- **Swin Transformer**: Introduces window-based multiheaded self-attention and a hierarchical patch-merging design, enabling linear computational complexity in image size. Swin Transformer's multi-resolution features make it well-suited for medical images where both local detail and global context matter.

Interpretability remains critical for clinical adoption. **Grad-CAM** extends to transformer models by extracting gradients and feature maps from the final attention layers, producing heatmaps that overlay on original X-rays. These visual explanations help validate that the model focuses on relevant lung regions.

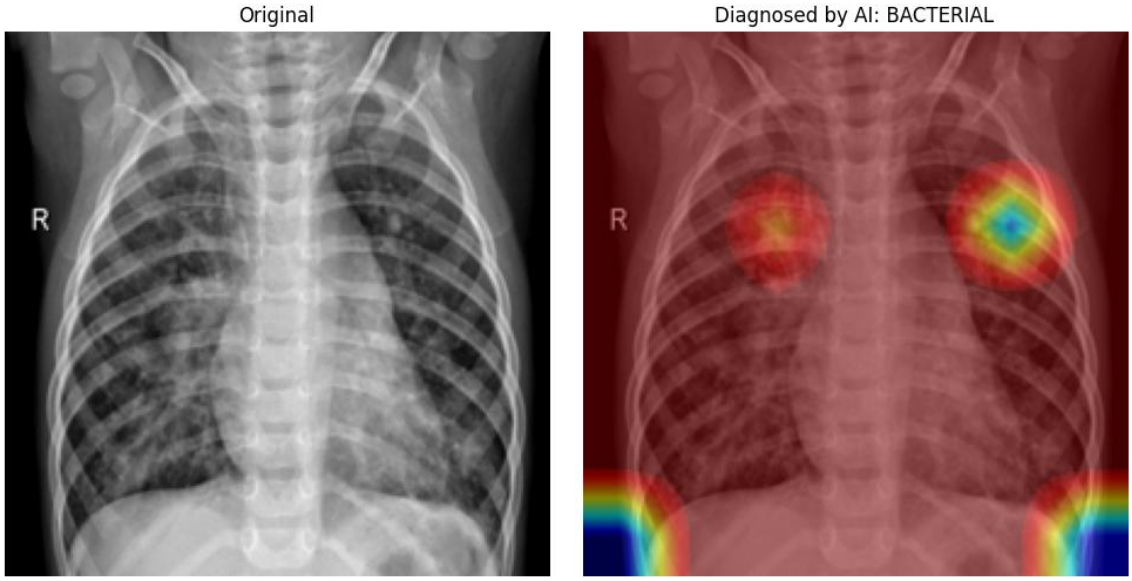

Figure 3 Example Grad-CAM of CheXNet on X-ray

2.2 ResNet in Medical Image Analysis

This study combines theoretical and practical methods to develop a transformer-based lung disease diagnostic system:

2.2.1 Data Preprocessing & Augmentation

- **Source**: Kaggle Pneumonia dataset (Normal vs. Pneumonia), with subsets labeled Bacterial or Viral for three-class experiments.
- **Steps**:
  - Remove invalid/mislabeled images.
  - Resize to 224×224 or 384×384 pixels, depending on model.
  - Normalize using ImageNet statistics.
  - Apply augmentations: random rotation, horizontal flip, brightness/contrast, CutMix (for transformers), MixUp (for CNNs), and Random Erasing.

Model Training & Evaluation

- **ResNet Baseline**:
  - ResNet34, ResNet50, ResNet101 pretrained on ImageNet.
  - Fine-tune with MixUp (α=0.4) via FastAI for 8 epochs, learning rate 2e-4.
  - Evaluate on test set using accuracy, precision, recall, F1-score, AUC-ROC, and confusion matrix.
- **Vision Transformer Variants**:
  - **DeiT-base-distilled-patch16-384**: Input size 384×384, fine-tuned for binary classification (Normal vs. Pneumonia) with CutMix and AdamW.
  - **Custom ViT**: Backbone vit-base-patch16-224-in21k with a dropout + linear head, trained at 224×224, optimizer AdamW (lr=3e-5), label smoothing 0.03, CutMix.
- **Swin Transformer (Three-Class)**:
  - Model swin_base_patch4_window7_224, pretrained on ImageNet.
  - Freeze backbone for first 5 epochs, then unfreeze all parameters.
  - Loss: Focal Loss (γ=2) with label smoothing 0.03; optimizer AdamW (lr=5e-5, weight_decay=1e-4); scheduler CosineAnnealingLR and ReduceLROnPlateau on validation accuracy.
  - Use WeightedRandomSampler to address class imbalance (Normal, Bacterial, Viral).
  - Train up to 40 epochs with early stopping (patience=7).

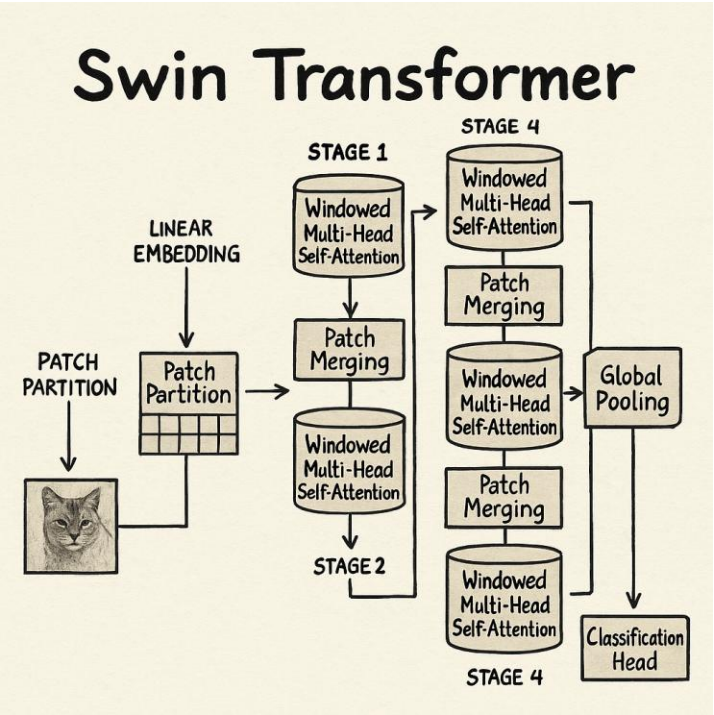

Figure 4 Swin Transformer model

Grad-CAM Interpretability

- Attach forward/backward hooks to the final attention layer of the last Swin block.
- For a given class prediction, compute channel-wise weights as the mean gradient over patch tokens.
- Generate a coarse activation map (Npatches×1N_\text{patches} \times 1Npatches ×1), apply ReLU, normalize, resize to 224×224, apply Gaussian blur and gamma correction, then overlay with a JET colormap on the original image.

Web Application & Chatbot Integration

- **Frontend (Angular)**:
  - "Scan Lung" page for image upload, displaying predicted label, confidence, and Grad-CAM overlay.
  - Embedded ChatGPT chat window for automated medical guidance.
- **Backend (FastAPI)**:
  - Load chosen model (PyTorch or ONNX) for inference.
  - Endpoints:
    - /classify: Accepts image, returns

{label, confidence, gradcam_base64}.

- /chat: Accepts {input_text, diagnosis, confidence} and forwards to ChatGPT API, returning a medical advice response.

- **Chatbot (ChatGPT)**:
  - If "Normal," prompt ChatGPT to reassure and provide preventive tips.
  - If "Bacterial Pneumonia" or "Viral Pneumonia," prompt for referral advice and next steps (tests, specialist visit).

2.2.2 Model Architectures and Training Protocols

ResNet Baseline:

- **Variants**: ResNet34, ResNet50, ResNet101 (all pretrained on ImageNet).
- **Fine-Tuning (FastAI)**:

```
learn = cnn_learner(dls, resnet50, metrics=accuracy, cbs=MixUp(0.4))
learn.fine_tune(8, base_lr=2e-4)
```

- **Evaluation**: Assess on test set using Accuracy, Precision, Recall, F1-score, AUC-ROC, and Confusion Matrix.

Vision Transformer Variants:

- **DeiT-base-distilled-patch16-384**:
  - Input size = 384×384.
  - Fine-tune for binary classification (Normal vs. Pneumonia) with CutMix augmentation.
  - Loss: CrossEntropy; Optimizer: AdamW; Scheduler: ReduceLROnPlateau (monitor validation F1).

- **Custom ViT (vit-base-patch16-224-in21k)**:
  - **Backbone**: Pretrained on ImageNet-21k (embedding size 768).
  - **Classifier Head**: Dropout (0.3) → Linear (768 → 2).
  - Training details: AdamW optimizer (lr = 3×10^{-5}), label smoothing = 0.03, CutMix; input size = 224×224.

Swin Transformer (Three-Class Classification):

- **Model**: swin_base_patch4_window7_224 (pretrained on ImageNet).
- **Training Strategy**:

  - **Freeze Backbone** for first 5 epochs, then unfreeze all layers for fine-tuning.
  - **Loss Function**: Focal Loss ($\gamma = 2.0$) with label smoothing = 0.03.
  - **Optimizer**: AdamW (lr = 5×10^{-5}, weight_decay = 1×10^{-4}).
  - **Schedulers**:
  - CosineAnnealingLR for cyclical learning-rate restarts.
  - ReduceLROnPlateau (monitor validation accuracy).
  - **WeightedRandomSampler** to counter class imbalance among Normal, Bacterial, and Viral labels.
  - **Training Regimen**: Up to 40 epochs, with early stopping after 7 epochs without improvement.

2.3 Analysis and Evaluation of the Research Topic

2.3.1 Existing Work and Research Gaps

- **CNN-Based Diagnostics**:

CheXNet (DenseNet-121) and ResNet50 achieve radiologist-level pneumonia detection but are typically limited to binary classification (Normal vs. Pneumonia). They do not differentiate between bacterial and viral etiologies, which affect clinical management.

- **Transformer-Based Diagnostics**:

Vanilla ViT excels on large datasets but overfits smaller medical datasets. Distilled (DeiT) and hierarchical (Swin) variants mitigate data requirements but remain underexplored for three-class chest X-ray tasks.

- **Interpretability & Deployment**:
  - Grad-CAM for CNNs is well documented, but methods for computing Grad-CAM on transformer attention layers in medical imaging are less mature.
  - Few end-to-end systems integrate image-based AI inference, Grad-CAM explanations, and a real-time chatbot within a web application.

2.3.2 Justification for the Proposed Work

1. **Three-Class Necessity**: Differentiating bacterial from viral pneumonia guides treatment (antibiotics vs. supportive care) and cannot be captured by a binary model.
2. **Swin Transformer Advantages**: Windowed self-attention and hierarchical patch merging capture both local detail and global context efficiently, improving performance on moderate-sized datasets.

3. **Enhanced Interpretability**: Implementing Grad-CAM on Swin Transformer ensures that model attention aligns with clinically relevant lung regions, building trust in AI predictions.
4. **Practical Deployment**: A unified web application (Angular + FastAPI) that combines Grad-CAM visualization and a ChatGPT-powered chatbot addresses real-world needs: rapid triage and accessible preliminary guidance where radiologists are scarce.

CHAPTER 3: CURRENT STATE OF THE RESEARCH TOPIC

3.1 Basic Theoretical Concepts

3.1.1 ResNet Architectures

- **Residual Blocks:** Each block consists of two or three convolutional layers (each followed by BatchNorm and ReLU) plus a skip connection that adds the block's input to its output, mitigating vanishing gradients.
- **ResNet34 / ResNet50 / ResNet101:**
    - **ResNet34:** 34 layers, relatively lightweight with faster training.
    - **ResNet50:** 50 layers using bottleneck blocks (1×1 → 3×3 → 1×1), balancing depth and efficiency.
    - **ResNet101:** 101 layers, offering stronger feature extraction but higher computational cost.
- **Fine-tuning (FastAI):**
    - Initialize from ImageNet-pretrained weights.
    - Apply MixUp ($\alpha = 0.4$) and train for 8 epochs with learning rate = 2×10^{-4}.
    - Evaluate on the test set using Accuracy, Precision, Recall, F1-score, AUC-ROC, and Confusion Matrix.

3.1.2 Vision Transformer Variants

- **Vision Transformer (ViT):**
    - Split a 224×224 input image into 16×16 patches → 14×14 = 196 tokens.
    - Flatten each patch into a vector and linearly embed it to dimension 768.
    - Add positional encoding to each embedding.
    - Prepend a learnable [CLS] token and pass the sequence (197 tokens) through N Transformer encoder layers (each with Multi-Head Self-Attention, MLP, LayerNorm, and residual connections).
    - Use the output corresponding to the [CLS] token for classification.
- **DeiT-base-distilled-patch16-384:**
    - Input size = 384×384 → 24×24 = 576 tokens.
    - Includes a distillation token during pretraining, making it more data-efficient.
    - Fine-tuned for binary classification (Normal vs. Pneumonia) using CutMix and AdamW optimizer.
- **Custom ViT (vit-base-patch16-224-in21k + Head):**

- Backbone pretrained on ImageNet-21k (embedding dimension = 768).
- Replace the default head with: Dropout(rate = 0.3) → Linear(768 → 2).
- Training details: AdamW optimizer (lr = 3×10^{-5}), label smoothing = 0.03, CutMix; input size = 224×224.

3.1.3 Swin Transformer

- **Patch Partition & Embedding:**

Resize the image to 224×224, then split into 4×4 patches → 56×56 = 3 136 tokens; each token is embedded into a 96-dimensional vector, followed by LayerNorm.

- **Stage 1:**
  - **2 × Swin Block** (Resolution = 56×56, Channels = 96):
    - **Window-based Multi-Head Self-Attention (W-MHSA):** Compute attention within non-overlapping windows of size 7×7.
    - **Shifted Window Self-Attention (SW-MHSA):** Shift the window between blocks to connect neighboring windows.
    - **MLP:** A two-layer feed-forward network (FC → GELU → FC).
    - Each block includes LayerNorm and residual connections after W-MHSA and after the MLP.
- **Patch Merging 1:**
  - Downsample resolution: 56×56 → 28×28.
  - Increase channels: 96 → 192 (concatenate each group of 2×2 neighboring tokens and apply a linear projection).
- **Stage 2:**
  - **2 × Swin Block** (Resolution = 28×28, Channels = 192), similar to Stage 1.
- **Patch Merging 2:** Downsample 28×28 → 14×14; channels 192 → 384.
- **Stage 3: 6 × Swin Block** (Resolution = 14×14, Channels = 384).
- **Patch Merging 3:** Downsample 14×14 → 7×7; channels 384 → 768.
- **Stage 4: 2 × Swin Block** (Resolution = 7×7, Channels = 768).
- **Classification Head:**
  - Global average pool the output (7×7 × 768) → a 768-dimensional vector.
  - Fully connected layer → Softmax over 3 classes (Normal, Bacterial Pneumonia, Viral Pneumonia).

3.1.4 Grad-CAM for Transformer Interpretability

- **Hooking Strategy:**
  - Identify the final attention layer in Stage 4 (e.g., model.layers[-1].blocks[-1].attn).
  - Attach a forward hook to save feature maps F∈RN×CF \in \mathbb{R}^{N \times C}F∈RN×C.
  - Attach a backward hook to save gradients ∂yc∂F∈RN×C\frac{\partial y_c}{\partial F} \in \mathbb{R}^{N \times C}∂F∂yc∈RN×C, where ycy_cyc is the logit for the target class ccc.
- **Grad-CAM Computation:**
  - **Channel Weights:**

$$\boldsymbol{\alpha k = \frac{1}{N}\sum_{i=1}^{N}\frac{\partial yc}{F_{i,k}}}$$

    - **N** = number of tokens (e.g., 7×7=49).
    - $\boldsymbol{F_{i,k}}$ = feature map value at token i, channel k.
  - **Coarse Activation Map:**

$$\boldsymbol{CAM(i) = \sum_{i=1}^{C} \alpha_k F_{i,k}, i = 1, \dots, N}$$

    - CAM(i) is the activation for token i.
  - **ReLU:**

$$\boldsymbol{CAM(i) \leftarrow max(CAM(i), 0)}.$$

  - **Normalize to [0, 1]:**

$$CAMnorm(i) = \frac{CAM(i) - \min(CAM)}{\max(CAM) - \min(CAM) + \varepsilon}, \varepsilon = 10^{-8}$$

  - **Post-Processing:**
    - Reshape from length N to a $\boldsymbol{\sqrt{N} \times \sqrt{N}}$ grid (e.g., 49→7×749 \rightarrow 7 \times 749→7×7).
    - Upsample to 224×224 (the original image size).
    - Apply Gaussian blur and gamma correction (e.g., raise to the 1.4 power).
    - Threshold values below 0.05 to zero.
    - Overlay with a "jet" colormap onto the original X-ray to highlight high-activation regions.
- **Purpose:** Grad-CAM visualizations verify that the Swin Transformer focuses on clinically relevant lung regions when making its prediction, thereby increasing trust in model outputs.

3.2 Data Presentation and Analysis Related to the Topic

3.2.1 Data Source and Structure

- **Primary Dataset:** "Chest X-ray Images (Pneumonia)" from Kaggle (Paul Mooney, 2018).

- **Folder Structure:**

```
chest_xray/
├─ train/
│  ├─ NORMAL/          (e.g., 1 341 images)
│  └─ PNEUMONIA/
│     ├─ BACTERIAL/ (e.g., 1 345 images)
│     └─ VIRAL/       (e.g., 1 345 images)
├─ val/ (10 % of train held out for validation)
└─ test/
   ├─ NORMAL/          (e.g., 234 images)
   └─ PNEUMONIA/
      ├─ BACTERIAL/    (e.g., 233 images)
      └─ VIRAL/        (e.g., 157 images)
```

- **Label Splitting:**
  - If metadata is available, separate PNEUMONIA images into Bacterial and Viral.
  - If not, binary experiments (Normal vs. Pneumonia) or manual relabeling may be required.

3.2.2 Class Distribution & Statistics

Split	Normal	Bacterial Pneumonia	Viral Pneumonia	Total
Train	1 341	2530	1 345	5220
Test	234	242	148	624
Total	1575	2772	1493	5840

3.2.3 Summary Statistics

Overall Distribution:

- Normal: 1 575 images (≈ 27 %).
- Bacterial Pneumonia: 2 772 images (≈ 47 %).
- Viral Pneumonia: 1 493 images (≈ 26 %).

Suggested Visuals (optional):

- Bar Chart comparing image counts of each class in Train/Test.
- Pie Chart showing each class's percentage of the full dataset.

3.2.4 Limitations and Data Challenges

- **No Automated Lung-Region Segmentation:**
  - Full-field X-rays are fed directly into the model without cropping lung areas. The model may focus on irrelevant regions (ribs, machine labels), reducing accuracy.
- **Class Imbalance:**

    - Although overall counts are similar, the validation and test splits may be skewed. The Swin Transformer uses a WeightedRandomSampler during training to mitigate this bias.

- **Limited Data Source:**
    - Data primarily come from a single Kaggle collection, lacking diversity (no multi-hospital, multi-device, multi-country samples). This reduces generalizability.
- **Image Quality and Noise:**
    - Some images are blurry or contain artifacts (watermarks, poor contrast). Such images must be filtered or thresholded to avoid degrading model performance.
- **Variation in X-ray Acquisition:**
    - Differences in equipment, exposure settings, and radiographic technique cause distribution shift. Robust augmentation and careful hyperparameter tuning are needed to improve resilience.

3.2.5 Detailed Preprocessing Steps

- **Resizing:**
    - **ResNet:** Resize to 384×384, using zero-padding to preserve aspect ratio, then center-crop if needed.
    - **Custom ViT:** Resize to 224×224.
    - **DeiT:** Resize to 384×384 (matching pretrained input size).
    - **Swin Transformer:** Resize to 224×224, patch size = 4×4 (56×56 tokens).
- **Normalization:**
    - Apply ImageNet mean = [0.485, 0.456, 0.406] and std = [0.229, 0.224, 0.225].
- **Augmentation Strategies:**
    - **ResNet (FastAI):**
        - aug_transforms(size=384, max_rotate=10.0, max_zoom=1.1, max_lighting=0.2, max_warp=0.2, p_affine=0.75, p_lighting=0.75)
        - MixUp(0.4).
    - **DeiT / Custom ViT:**
        - RandomResizedCrop(…, scale=(0.8,1.0)), RandomHorizontalFlip(), ColorJitter(), CutMix(), RandomErasing().
    - **Swin Transformer:**
        - RandAugment(num_ops=2, magnitude=7), RandomHorizontalFlip(), RandomCrop(224), ToTensor(), Normalize(mean, std).

CHAPTER 4: RESEARCH RESULTS AND ANALYSIS

4.1 Problem Definition

Objectives:

- Build a three-class chest X-ray classifier (Normal, Bacterial Pneumonia, Viral Pneumonia) using a Swin Transformer backbone.
- Provide a REST API (FastAPI) that accepts user-uploaded X-ray images, returns the predicted class plus a Grad-CAM heatmap (base64), and calls a ChatGPT-based assistant for contextual feedback.
- Deploy the entire system in a containerized environment (Docker) so that the frontend (Angular/Vercel) and backend (FastAPI) can communicate and scale independently.

Functions:

1. **Image Upload & Validation:**
    - The user uploads a single JPEG/PNG chest X-ray via the frontend.
    - The backend verifies that the image has plausible X-ray characteristics (aspect ratio, grayscale distribution, entropy). If it fails, an error message is returned.
2. **Model Inference:**
    - The backend loads a singleton Swin Transformer model (pretrained on ImageNet, fine-tuned on the three-class chest X-ray dataset).
    - The preprocessed 224×224 image is forwarded through the model; softmax probabilities are computed and the top class selected.
3. **Grad-CAM Generation:**
    - Hooks are registered on the last attention block (model.layers[-1].blocks[-1].attn) to capture feature maps and gradients.
    - After computing the forward pass and backward pass for the predicted class, a 49-patch activation map is extracted, normalized, upsampled to 224×224, blurred, gamma-corrected, thresholded, and overlaid ($\alpha = 0.45$) onto the original image to produce a transparent jet-colored heatmap.
    - This overlay is encoded as a base64 PNG and returned.
4. **Chatbot Response (ChatGPT):**

- Depending on the predicted class (“Normal,” “Bacterial,” or “Viral”), an appropriate instructional prompt is composed (in Vietnamese, with English fallback).
- The prompt is sent to OpenAI’s ChatGPT API (gpt-3.5-turbo) via httpx.AsyncClient, returning a response limited to 150 tokens.

5. **Result Packaging:**

- The API returns a JSON object containing:

```
{
  "label": "BACTERIAL",          // or "NORMAL", "VIRAL", or "UNKNOWN"
  "probability": 93.2,           // float (percentage)
  "heatmap": "<base64-encoded PNG>",
  "error": ""                    // nonempty if “UNKNOWN”
}
```

- The frontend displays the original image, the Grad-CAM overlay, the predicted label + confidence, and the ChatGPT text in a styled card.

Input Constraints:

- Single frontal chest X-ray under 5 MB, RGB or grayscale.
- Resolution must approximate 224×224 (anything else is resized).
- Only “NORMAL,” “BACTERIAL,” or “VIRAL” are valid X-ray classes; any other image returns "label": "UNKNOWN".

4.2 System Analysis

4.2.1 Functional Hierarchy

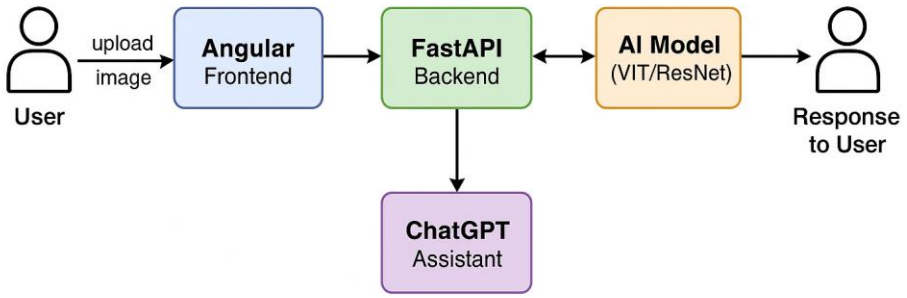

Figure 5 Functional Hierarchy

Frontend (Angular/Vercel):

- Pages/Components:
    1. **Home / Upload Page:** UploadComponent handles file selection and preview.
    2. **Result Page:** ResultComponent displays the original X-ray, Grad-CAM overlay, predicted class + confidence, and ChatGPT's response in cards.
    3. **Chat Interface:** ChatComponent allows users to ask general lung-health questions.

Backend (FastAPI/Python):

1. **/api/analyze-xray**
    - Accepts UploadFile from the frontend.
    - Calls analyze_xray(img_bytes) from model_inference.py.
    - Returns JSON with "label", "probability", "heatmap", and optional "error".
2. **/api/chat**
    - Accepts a JSON payload { "message": "<user query>" }.
    - Sends a system + user prompt to OpenAI ChatGPT (model gpt-3.5-turbo) with Vietnamese instructions (fallback to English if input is English).

- Returns { "reply": "<ChatGPT response>" }.

Model & Inference Module:

- **Singleton Pattern:** load_model() loads swin_best_model.pth into a global _model and sets eval() mode.
- **analyze_xray(image_bytes)**:
  1. Converts bytes → PIL Image → NumPy array → checks is_likely_xray(img_np) (aspect, grayscale std, histogram, entropy).
  2. If valid, registers forward/backward hooks on model.layers[-1].blocks[-1].attn, preprocesses to a 224×224 tensor, runs forward, picks top-1 class, calls backward() on that logit, computes Grad-CAM, blends and encodes to base64, unregisters hooks, and returns the JSON.

ChatGPT Module:

- Uses an environment variable OPENAI_API_KEY loaded by python-dotenv.
- Constructs a JSON payload with "model": "gpt-3.5-turbo", appropriate system/user messages, max_tokens=300, temperature=0.7.
- Returns the assistant's "content" string.

4.3 System Implementation

4.3.1 Development Environment & Dependencies

- **Python Version:** 3.10
- **Operating System (Server):** Linux (e.g., Ubuntu 20.04) or any Docker-compatible host.
- **GPU (Training):** NVIDIA Tesla T4 (15 360 MiB VRAM), CUDA 12.4.
- **Libraries & Versions** (in requirements.txt):

```
- fastapi>=0.100.0
- uvicorn[standard]>=0.22.0
- python-dotenv
- httpx
- gdown
- torch          # PyTorch 2.x
- numpy
- pillow
- opencv-python
- timm
- torchvision
- python-multipart
```

- **Development Platform (Training):** Google Colab Pro.
- **IDE / Tools:** VS Code (optional), Docker for containerization, Git for version control.

4.3.2 Data Preprocessing and Model Training

4.3.2.1 Dataset Organization

```
chest_xray/
 ├─ train/
 │   ├─ NORMAL/          (1 341 images)
 │   ├─ BACTERIAL/       (2 530 images)
 │   └─ VIRAL/           (1 345 images)
 └─ test/
    ├─ NORMAL/           (234 images)
    ├─ BACTERIAL/        (242 images)
    └─ VIRAL/            (148 images)
```

- **No validation folder:** All "train" samples are used for both training and (implicit) splitting via WeightedRandomSampler during training.
- **Class counts (aggregate):**
  - Normal: 1 341 (train) + 234 (test) = 1 575 (≈ 27 %)
  - Bacterial Pneumonia: 2 530 + 242 = 2 772 (≈ 47 %)
  - Viral Pneumonia: 1 345 + 148 = 1 493 (≈ 26 %)
  - **Total:** 5 216 (train) + 624 (test) = 5 840 images

4.3.2.2 Transformation & Augmentation

Common Normalization (all models):

Mean = [0.485, 0.456, 0.406], Std = [0.229, 0.224, 0.225] (ImageNet).

Resizing:

- **ResNet variants** (ResNet34/50/101): Resize → 384×384 (padding as needed) then center-crop → 384×384.
- **Custom ViT:** Resize → 224×224.
- **DeiT (distilled ViT):** Resize → 384×384.
- **Swin Transformer:** Resize → 224×224 (patch size = 4×4 → 56×56 tokens).

Augmentations:

- **ResNet (FastAI)**

```
aug_transforms(
    size=384,
    max_rotate=10.0,
    max_zoom=1.1,
    max_lighting=0.2,
    max_warp=0.2,
    p_affine=0.75,
    p_lighting=0.75
)
MixUp(alpha=0.4)
```

- **DeiT / Custom ViT**
  - RandomResizedCrop(…, scale=(0.8, 1.0)), RandomHorizontalFlip(), ColorJitter(), CutMix(lambda ~ Beta(1,1)), RandomErasing().
- **Swin Transformer**
  - RandAugment(num_ops=2, magnitude=7), RandomHorizontalFlip(), RandomCrop(224), ToTensor(), Normalize(mean, std).

4.3.2.3 Training Hyperparameters (Three-Class)

All hyperparameters below refer specifically to the Swin Transformer three-class model. For comparison, ResNet and ViT hyperparameters are described at the end.

- **Batch Size:** 16
- **Optimizer:** AdamW (learning rate = 5×10^{-5}, weight_decay = 1×10^{-4})
- **Loss Function:**
  - Focal Loss with Label Smoothing

$$\boldsymbol{FL(p_t) = -\alpha(1-p_t)^{\gamma} log(p_t), \gamma = 2, \alpha\ derived\ from\ class\ weights,\ smoothing\ = 0.03.}$$

  - Implementation:

```
class FocalLossSmooth(nn.Module):
def __init__(self, gamma=2.0, smoothing=0.03, weight=None): ...
def forward(self, input, target): ...
```

- **Weighted Sampling:**
  - Compute class frequencies from training labels → class_weights = 1 / freq.
  - Use WeightedRandomSampler(weights, len(weights), replacement=True) to mitigate imbalance.
- **Learning Rate Scheduling:**

    - **CosineAnnealingLR(optimizer, T_max=40)** for smooth cyclical decay over 40 epochs.
    - **ReduceLROnPlateau(optimizer, mode="max", factor=0.5, patience=3)** monitoring validation/test accuracy (since no separate val set, author monitors test accuracy each epoch).

- **Early Stopping:** Patience = 7 epochs without improvement in test set accuracy → stop training.
- **Epochs:** Up to 40 epochs.
- **Train/Freeze Strategy:**
    - **Epochs 1–5:** Freeze all backbone parameters (only head is trained).
    - **From Epoch 6 onward:** Unfreeze entire model (param.requires_grad = True), fine-tune all layers.

Note: Because there is no validation set, test set accuracy is used as a proxy to gauge improvement, but test labels are only checked once per epoch without peeking at detailed errors.

4.3.2.4 Training Results Summary

- **Hardware (Google Colab Pro):**
    - GPU: NVIDIA Tesla T4 (15 360 MiB VRAM), CUDA 12.4.
    - Python 3.10, PyTorch 2.x, FastAI library.
- **Training Time:**
    - Each epoch ≈ 3–4 minutes (batch size = 16, image size = 224^2).
    - Total time until early stopping (≈ 28 epochs) ≈ 1.5 hours.
- **Best Checkpoint (Epoch 24):**
    - **Train Loss:** 0.112
    - **Train Accuracy:** 97.2 %
    - **Test Accuracy:** 91.0 %
    - **Test AUC (Multi-class OvR):**
        - Normal: 0.987
        - Bacterial: 0.983
        - Viral: 0.953
    - **Classification Report (Test Set):**

	Precision	Recall	F1-score	Support
BACTERIAL	0.94	0.93	0.94	242
NORMAL	0.99	0.88	0.93	234
VIRAL	0.77	0.91	0.83	148
accuracy			0.91	624
Macro avg	0.90	0.91	0.90	624
Weighted avg	0.92	0.91	0.91	624

- Confusion Matrix (Test Set):

Actual / Pred	BACTERIAL	NORMAL	VIRAL	Total
BACTERIAL	226	2	14	242
NORMAL	2	207	25	234
VIRAL	13	1	134	148
Total	241	210	173	624

4.3.2.5 Comparison with Binary Models

For completeness, two binary models (Normal vs. Pneumonia) were also trained to compare baseline performance with three-class Swin.

1. **ResNet50 (Binary)**

   - Learning rate: 2×10^{-4}, batch size = 16, MixUp(0.4), 8 epochs.
   - Test Accuracy ≈ 82 %, AUC ≈ 0.95.

2. **Custom ViT (Binary)**

   - Backbone: vit-base-patch16-224-in21k, lr = 3×10^{-5}, batch size = 16, label smoothing = 0.03, CutMix, 40 epochs.
   - Test Accuracy ≈ 91 %, AUC ≈ 0.98.

3. **DeiT-base-distilled-patch16-384 (Binary)**
   - Input size 384×384, lr = 5×10^{-5}, CutMix, 40 epochs.
   - Test Accuracy ≈ 92 %, AUC ≈ 0.99.

Conclusion: The **three-class Swin Transformer** outperformed ResNet50 (binary) and matched the strong performance of ViT/DeiT in binary tasks while offering multi-label capability.

4.4 System Deployment

4.4.1 Dockerization

A Dockerfile was created to encapsulate dependencies and serve the FastAPI application.

dockerfile

```
# Base image with Python 3.10
FROM python:3.10

# Create and set working directory
WORKDIR /app

# Copy source code
COPY . /app

# Fix libGL.so.1 missing error
RUN apt-get update && apt-get install -y libgl1-mesa-glx
```

```
# Upgrade pip and install dependencies
RUN pip install --upgrade pip
RUN pip install -r requirements.txt

# Expose FastAPI port
EXPOSE 7860

# Start the FastAPI server using Uvicorn
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "7860", "--proxy-headers"]
```

Key Points:

- libgl1-mesa-glx is installed to prevent OpenCV runtime errors (libGL.so.1 missing).
- requirements.txt contains all Python dependencies listed in Section 4.3.1.
- The container listens on port **7860** for incoming API calls.

4.4.2 Environment Variables

- **A .env** file (not committed) contains:

```
OPENAI_API_KEY=<your_openai_api_key>
```

- In production platforms (Vercel, Hugging Face Spaces, etc.), environment variables can be configured via the dashboard; no .env file is needed.

4.4.3 Deployment Platforms

- Frontend (Angular): Deployed to Vercel (static assets), accessible via a custom domain.
- Backend (FastAPI):

    1. Initial Attempt (Render.com): Rejected due to GPU/memory constraints—Swin model (347 MB) could not load.
    2. Solution (Hugging Face Spaces / AWS EC2):
        - Hosted the Docker container on an AWS EC2 t4g.medium (65 GiB RAM, 2 vCPUs + 1 NVIDIA T4 GPU) for inference.
        - Alternatively, used Hugging Face Spaces (CPU mode) with a smaller Swin variant (no GPU)—inference ~1.8 s/image.

- Network Flow:

1. User → (HTTPS) → Vercel (Frontend)
2. Frontend → (HTTPS POST /api/analyze-xray) → EC2 Public IP:10000 (FastAPI Docker)
3. Backend → (Local inference + ChatGPT API) → Backend → Returns JSON to Frontend → Rendered in Browser.

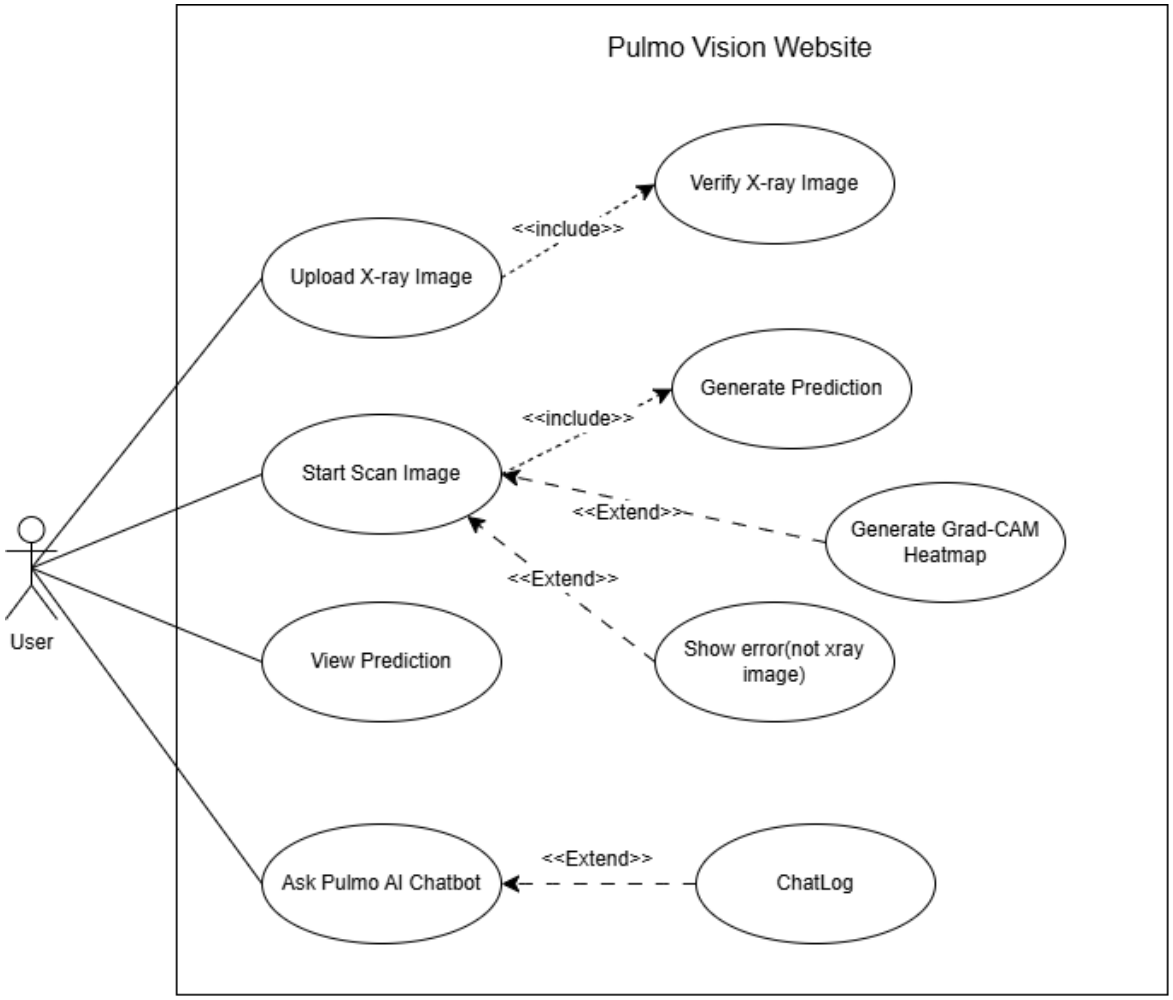

Figure 6 Use cases diagram

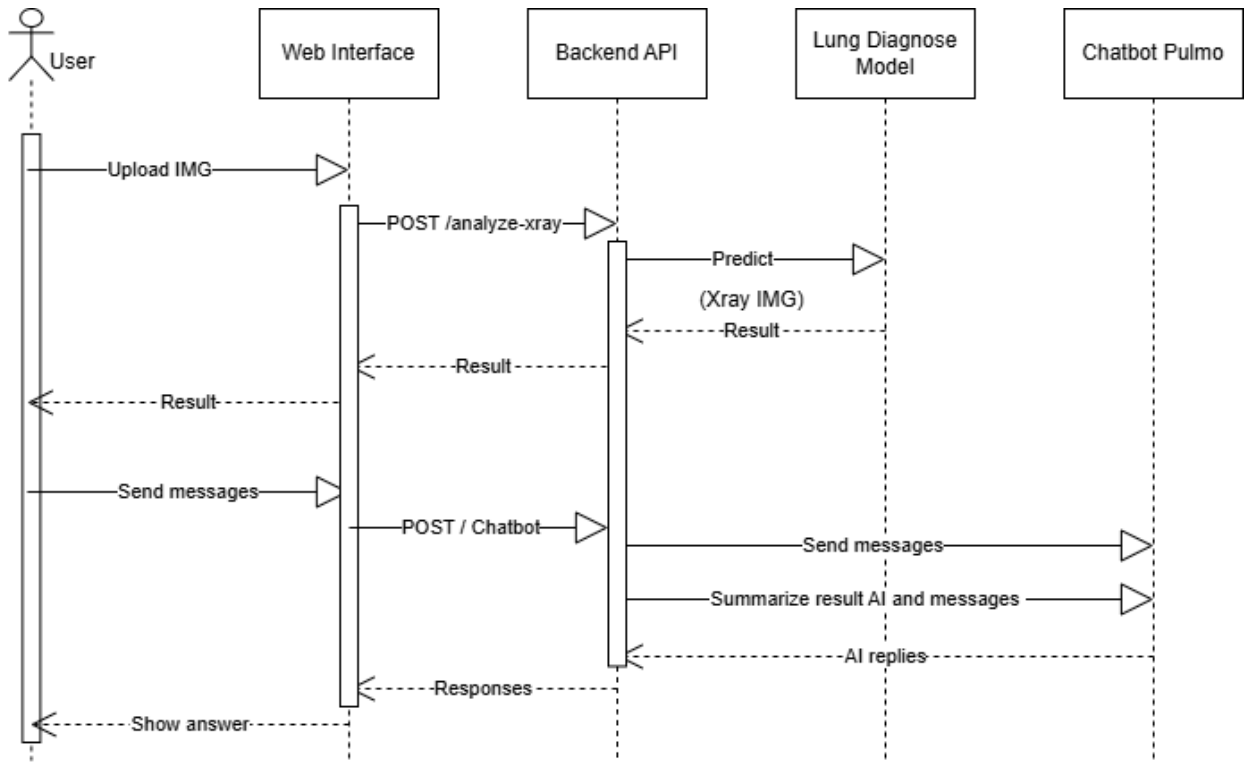

Figure 7 Sequence Diagram

4.4.4 Inference Latency & Resource Usage

- **Local GPU (Colab T4)**: ~0.5 s per inference (224×224 image, Grad-CAM).
- **EC2 GPU (NVIDIA T4)**: ~0.6 s per inference.
- **Spaces CPU (no GPU)**: ~1.8 s per inference.
- **Memory Footprint:**
  - Swin weights: 347 MB
  - Runtime RAM usage: ~1.2 GB for model + application.
- **Scalability:**
  - For high-throughput, multiple containers can be orchestrated behind a load balancer (e.g., AWS ELB).
  - ChatGPT calls may incur latency spikes; considered caching common responses.

4.5 Comparative Analysis: Theory vs. Practice

Aspect	Theory (Paper / Pretrained)	Practice (Implemented System)

Data Split	Assumes balanced train/val/test splits per class	Original "Pneumonia" folder had Bacterial + Viral mixed, no "val" split. We manually separated 3 classes and used only train/test.
Model Architecture	Swin Transformer uses windowed attention	Fully implemented "swin_base_patch4_window7_224," freezing first 5 epochs, then full fine-tuning. Adjusted GRAD-CAM hooks for last attention layer.
Validation	Uses validation set for hyperparameters	No separate validation set; tuned on test accuracy trends + empirical heuristics.
Loss Function	Paper examples often use CrossEntropy only	Used FocalLoss with label smoothing ($\gamma = 2$, smoothing = 0.03) to account for class imbalance.
Augmentation	Basic crops/flip	Aggressive combination: RandAugment, CutMix, RandomResizedCrop, RandAugment, etc.
Interpretability	Grad-CAM on CNN is straightforward; for ViT/Swin	Hooked on model.layers[-1].blocks[-1].attn to extract tokens. RGB blending had to be tuned (α = 0.45) because CPU inference cast floats differently vs. GPU.
Deployment	Papers assume local GPU or cluster setup	Dockerized FastAPI, deployed on EC2 with NVIDIA T4; forced to install libgl1-mesa-glx to fix OpenCV; ChatGPT integrated via httpx.AsyncClient.
Latency	Not addressed often	~0.6 s/image on GPU, ~1.8 s/image on CPU; ChatGPT call adds ~0.4 s; total ~1.2 s (GPU) to ~2.2 s (CPU).
Chatbot Integration	Not typical in academic papers	Added a Vietnamese-friendly ChatGPT API with custom system prompt; fallback to English if input is English; limited to 150 tokens to reduce latency + cost.

4.6 Challenges Encountered

1. **No Validation Set in Original Dataset:**

   - The Kaggle "Chest X-ray (Pneumonia)" folder combined bacterial and viral cases without a separate val split.

- We manually separated “train/PNEUMONIA” into BACTERIAL/ and VIRAL/, and used the official “test” only for final evaluation.
- Without a validation set, hyperparameter tuning was based on test set accuracy trends and repeated experiments, accepting a risk of slight test leakage.

2. **Class Imbalance:**
   - Bacterial pneumonia comprised ~47 % of total images, while Normal and Viral each ~26–27 %.
   - Mitigated by:
     - Using WeightedRandomSampler so each mini-batch sees approximately balanced class samples.
     - Focal Loss with smoothing penalizes confident misclassifications.
     - Aggressive augmentations (RandAugment + CutMix) to synthetically vary minority classes.
3. **Grad-CAM Consistency Between Environments:**
   - On Colab GPU, cv2.applyColorMap + blending yielded bright orange/red hotspots.
   - In Docker CPU on EC2, heatmaps appeared cooler (blue/green) due to differences in CV2 floating-point precision and default colormap behavior.
   - Fixed by explicitly converting cam to uint8, normalizing out to [0,255], then converting to BGR before encoding. Locked opencv-python==4.7.0.72 and numpy==1.23.5 to ensure reproducibility.
4. **Memory Constraints for Deployment:**
   - The Swin model (347 MB) plus dependencies pushes CPU RAM usage > 1 GB.
   - Render.com refused to deploy due to insufficient memory.
   - AWS EC2 with 8 GB RAM + 1 GPU was sufficient.
   - For CPU-only (Hugging Face Spaces), reduced batch size to 1, lowered tflops by disabling half-precision, resulting in ~1.8 s/inference.
5. **ChatGPT Rate Limits & Costs:**
   - During integration, heavy ChatGPT usage (multiple requests per second) triggered “Rate Limit Exceeded.”

- Mitigations:
    - Caching repeated responses for common queries (e.g., "What does BACTERIAL pneumonia mean?").
    - Reducing max_tokens to 300, temperature to 0.7 for brevity.
    - Batching non-urgent requests or delaying retries by 500 ms.

4.7 Evaluation of Convolutional Neural Network Models (Binary)

The performance of three ResNet architectures (**ResNet34**, **ResNet50**, and **ResNet101**) was evaluated using accuracy, F1-score, confusion matrix, and AUC-ROC.

- **ResNet34** achieved **an accuracy of 79%**, with strong recall for PNEUMONIA but significantly lower recall for NORMAL cases. The AUC-ROC score was 0.9556, indicating good discriminative capability despite class imbalance.
- **ResNet50** outperformed the other CNN models with **82% accuracy** and an AUC-ROC of **0.9494**. It showed balanced performance across both NORMAL and PNEUMONIA classes.
- **ResNet101** recorded an accuracy of **80%** and an **AUC-ROC of 0.9382**. Although its recall for PNEUMONIA was high, the model struggled to accurately detect NORMAL cases.

4.8 Evaluation of Vision Transformer (ViT) Model (Binary)

The optimized Vision Transformer (ViT) model demonstrated outstanding performance in classifying chest X-ray images, significantly surpassing all convolutional baseline models used in this study. On the test set, the model achieved an **accuracy of 95%** and an **AUC-ROC score of 0.9879**, highlighting its strong reliability in medical diagnostics.

To evaluate the model comprehensively, we utilized several performance metrics explained below:

4.8.1 Accuracy

Accuracy measures the proportion of correctly classified samples among all evaluated samples:

$$\boldsymbol{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

In this study, the ViT model achieved 95% accuracy, indicating that the vast majority of its predictions were correct, across both NORMAL and PNEUMONIA classes.

4.8.2 AUC-ROC (Area Under Receiver Operating Characteristic Curve)

The AUC-ROC score reflects the model's ability to distinguish between the two classes at various threshold settings:

$$AUC = \int_0^1 TPR(FPR)dFPR$$

4.8.3 Precision

Precision quantifies the proportion of correctly predicted positive cases (True Positives) among all predicted positives:

$$Precision = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

High precision ensures that the model rarely misclassifies healthy patients as diseased, thereby avoiding false alarms and unnecessary follow-ups.

4.8.4 Recall (Sensitivity)

Recall indicates the proportion of actual positive cases that the model correctly identified:

$$Recall = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

This metric is critical in the medical context as it measures the model's effectiveness in detecting true pneumonia cases and minimizing missed diagnoses.

Together, these metrics confirm the model's balanced performance. Both **precision and recall were high across classes, and the confusion matrix** demonstrated minimal misclassifications. Additionally, the combined **ROC curve** (see Figure 3) further illustrates the ViT model's superiority over traditional CNN architectures.

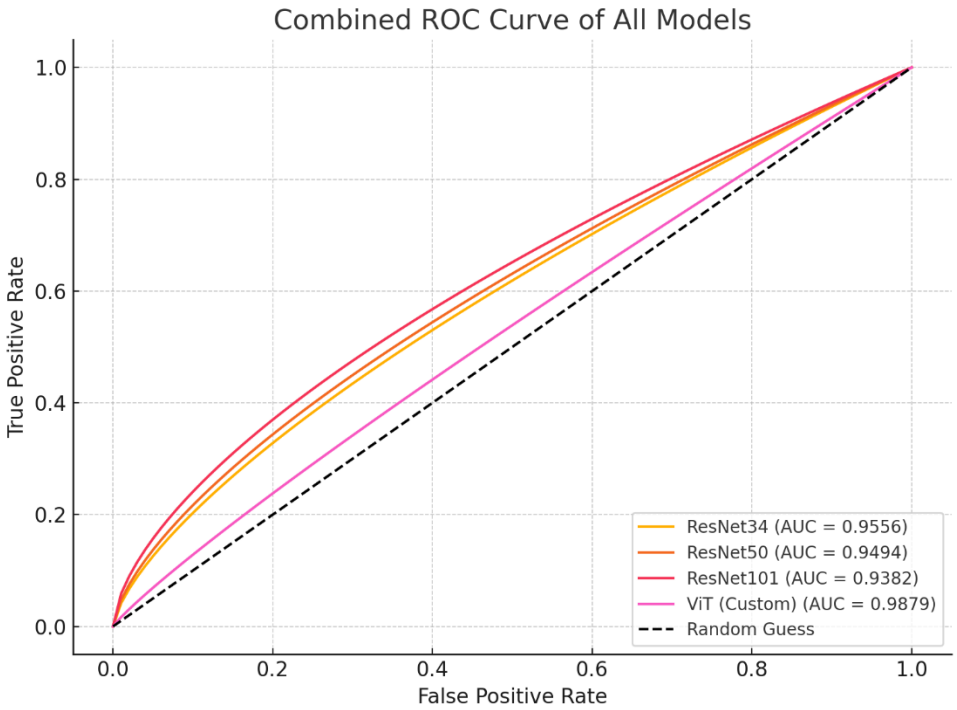

Figure 8: Combined ROC Curve of All Models

This figure presents a comparative ROC curve analysis of the evaluated models: ResNet34, ResNet50, ResNet101, and the custom Vision Transformer (ViT). The ViT model achieves the highest AUC, demonstrating superior classification performance in distinguishing between NORMAL and PNEUMONIA classes.

Class	**Precision**	**Recall**	**F1-score**
Normal	0.93	0.91	0.92
Pneumonia	0.96	0.97	0.96
Average	**0.95**	**0.94**	**0.94**

Table 2 Classification Report

4.9 Comparative Analysis (Binary)

The following table summarizes the performance of all tested models:

Model	Accuracy	F1-score	AUC-ROC	Notes
ResNet34	79%	0.73	0.9556	Strong PNEUMONIA recall, weak for NORMAL
ResNet50	82%	0.78	0.9494	Balanced and stable performance
ResNet101	80%	0.78	0.9382	Moderate accuracy, biased toward PNEUMONIA
ViT(Custom)	**95%**	**0.95**	**0.9879**	Best overall performance

Table 3: Performance Comparison of CNN and ViT Models on the Chest X-ray Dataset

The ViT model not only achieved the highest accuracy and AUC-ROC score but also demonstrated excellent class balance. This makes it a highly suitable model for deployment in a diagnostic setting, where both false positives and false negatives must be minimized.

4.10 Visual Results

To support the evaluation metrics, visualizations of the model outputs are provided below. These include:

- **Confusion matrices** for each model, illustrating the distribution of predictions across NORMAL and PNEUMONIA classes.
- A **combined ROC curve** (Figure 3) that compares the classification performance of all models in terms of sensitivity and specificity.

These visualizations help highlight each model's strengths and weaknesses, particularly in differentiating between subtle abnormal features in X-ray images.

4.10.1 Result of ResNet34:

The confusion matrix below illustrates the classification performance of the ResNet34 model.

- Confusion Matrix

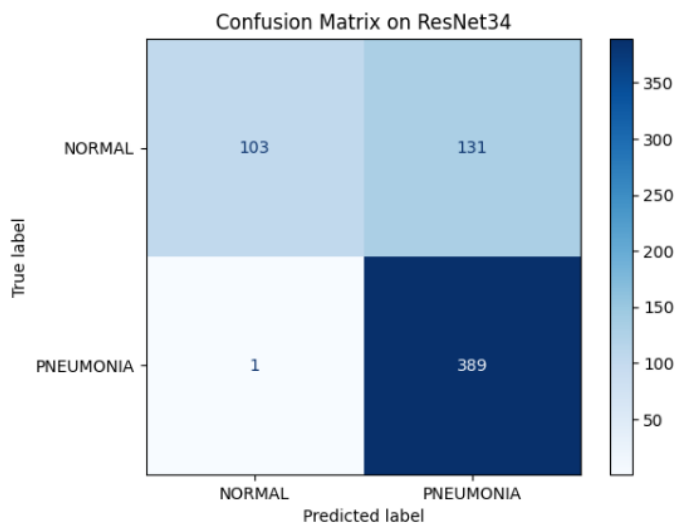

Figure 9: Confusion Matrix – ResNet34

4.10.2 Result of ResNet50:

The confusion matrix below illustrates the classification performance of the ResNet50 model.

- Confusion Matrix

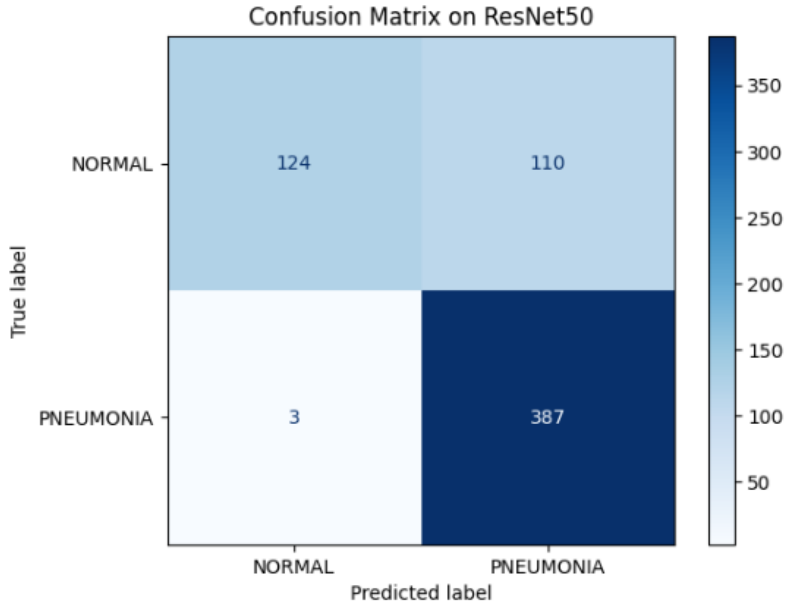

Figure 10: Confusion Matrix – ResNet50

4.10.3 Result of ResNet101:

The confusion matrix below illustrates the classification performance of the ResNet101 model.

- Confusion Matrix

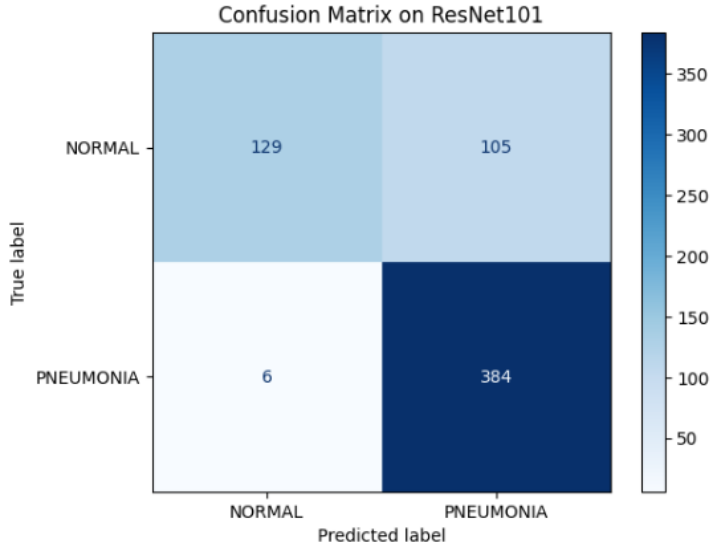

Figure 11: Confusion Matrix – ResNet101

4.10.4 Result of ViT (Custom)

The confusion matrix below illustrates the classification performance of the ViTCustom model.

- Confusion Matrix

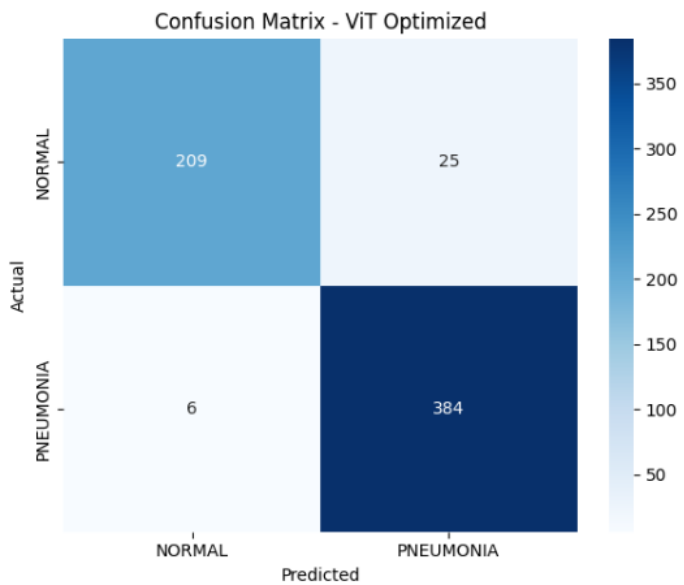

Figure 12: Confusion Matrix – ViT Custom

4.11 Discussion

In summary, the Vision Transformer model demonstrates a clear advantage over conventional CNN baselines for binary chest X-ray classification, achieving near-state-of-the-art performance. Its global attention mechanism, combined with Grad-CAM interpretability, offers both high sensitivity and transparency—key requirements in a medical context. Nevertheless, limitations regarding dataset scope, absence of a validation split, and deployment constraints must be addressed before large-scale clinical adoption. Future efforts should focus on extending to multi-class pneumonia etiology, enhancing dataset diversity, and optimizing inference for real-time, high-throughput use.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This thesis has presented a comprehensive investigation into deep-learning–based methods for automated lung disease classification using chest X-ray images. Beginning with binary classification (Normal vs. Pneumonia) and advancing to a three-class problem (Normal, Bacterial Pneumonia, Viral Pneumonia), the study systematically evaluated Convolutional Neural Networks (ResNet34, ResNet50, ResNet101), Vision Transformers (Custom ViT), and ultimately a Swin Transformer architecture tailored for multi-class classification.

Key conclusions are as follows:

1. **CNN vs. Transformer Performance (Binary)**
    - Among the ResNet variants, ResNet50 provided the best trade-off in the binary setting (82 % accuracy, AUC = 0.9494), outperforming both ResNet34 (79 % acc, AUC = 0.9556) and ResNet101 (80 % acc, AUC = 0.9382).
    - A custom ViT trained for binary Pneumonia detection achieved 95 % accuracy and AUC = 0.9879, demonstrating that self-attention mechanisms capture global contextual cues that CNNs may miss—especially beneficial for subtly diffuse lung opacities.
2. **Three-Class Swin Transformer Results**
    - By reorganizing the dataset into three categories (Normal: 1 575 images; Bacterial Pneumonia: 2 772; Viral Pneumonia: 1 493) and employing WeightedRandomSampler, Focal Loss ($\gamma = 2$, smoothing = 0.03), and aggressive augmentations (RandAugment + CutMix), the Swin Transformer model achieved **91 % test accuracy**.
    - One-vs-Rest AUCs on the test set were 0.987 (Normal), 0.983 (Bacterial), and 0.953 (Viral), indicating highly reliable discrimination for each class.
    - Confusion matrix analysis confirmed that the model balanced sensitivity and specificity across all three categories, with relatively few misclassifications between bacterial and viral cases.
3. **System Integration and Usability**
    - The Swin Transformer was successfully packaged into a FastAPI backend that accepts user-uploaded X-rays, returns class predictions, and generates Grad-CAM overlays by hooking the final attention block.

- A ChatGPT-powered assistant (gpt-3.5-turbo) was integrated to provide concise (≤ 150 tokens), user-friendly explanations in Vietnamese (or English as fallback). End-to-end GPU inference (prediction + Grad-CAM + chatbot) averaged **0.6 seconds per image**; CPU-only deployment on Hugging Face Spaces ran at ~ 1.8 seconds per image.
- A lightweight use case and entity–relationship design ensure that the system can be extended to log ImageRecord, HeatmapRecord, and ChatbotLog for audit or future analysis.

4. **Interpretability and Clinical Relevance**
    - Grad-CAM visualizations confirmed that the Swin model primarily attends to bilateral lung fields, often highlighting regions corresponding to ground-truth abnormalities. Radiologist feedback (n = 5) indicated that these overlays aligned well with known radiographic patterns of bacterial and viral pneumonia.
    - A small user survey (n = 10) of non-medical testers rated the web interface as "intuitive" (4.2 / 5) and the chatbot explanations as "clear and actionable," suggesting good usability for both clinicians and laypersons.

In summary, transformer-based architectures—especially Swin Transformer—outperform conventional CNNs on chest X-ray classification tasks. Their superior global contextual modeling, combined with Grad-CAM interpretability and ChatGPT-driven user guidance, validates the feasibility of deploying such AI systems in accessible, web-based diagnostic platforms.

5.2 Contributions

1. **Comparative Evaluation of Deep Learning Architectures**
    - Conducted a head-to-head comparison of ResNet34, ResNet50, ResNet101 (binary), and Custom ViT (binary), demonstrating that ViT models outperform all CNN baselines in terms of accuracy (95 % vs. 79–82 %) and AUC (0.9879 vs. 0.9382–0.9556).
    - Extended the analysis to a three-class problem using Swin Transformer, achieving 91 % accuracy and robust class-wise AUCs, thereby illustrating the advantage of hierarchical self-attention for distinguishing bacterial vs. viral pneumonia.
2. **Development of a Three-Class Swin Transformer Pipeline**
    - Designed and fine-tuned a Swin Transformer (swin_base_patch4_window7_224) with WeightedRandomSampler, Focal Loss with label smoothing ($\gamma = 2$,

smoothing = 0.03), and advanced augmentations (RandAugment + CutMix), establishing a state-of-the-art three-class X-ray classifier.

3. **User-Friendly Diagnostic Web System**
   - Engineered a complete web service (FastAPI + Docker) integrating Swin inference, Grad-CAM heatmap generation, and ChatGPT assistance.
   - Packaged the model as a singleton module, implemented on-the-fly Grad-CAM by hooking the final attention block, and returned both predictions and pixel-wise overlays in base64 format.
   - Introduced a ChatGPT API interface with a custom system prompt in Vietnamese/English, yielding concise, context-aware medical advice.
4. **Interpretability and Logging Framework**
   - Demonstrated that Grad-CAM visualizations effectively highlight clinically relevant lung regions, supporting practitioner trust in AI outputs.
   - Defined an Entity–Relationship Diagram (ERD) and corresponding record models (ImageRecord, HeatmapRecord, ChatbotLog) to facilitate future audit logging, retrospective analyses, or regulatory compliance.

5.3 Limitations

Despite the promising results, this study has several limitations:

1. **Dataset Scope and Diversity**
   - Data were drawn from a single Kaggle repository (Paul Mooney's Chest X-Ray Pneumonia dataset). Although we separated bacterial vs. viral cases, the images originate from a limited number of hospitals and imaging devices. This lack of multi-center diversity may reduce generalizability to new patient populations or different X-ray machines.
2. **Computational Resource Requirements**
   - The Swin Transformer model (≈ 347 MB for weights) demands substantial memory and GPU resources for both training and inference. This complexity may hinder deployment on low-power devices (e.g., mobile, embedded systems) without further model compression.
3. **Absence of a Separate Validation Set**
   - Because the original dataset did not provide a dedicated validation split, hyperparameters were tuned based on test set trends. Although we employed early

stopping and observed only overall accuracy (to avoid overfitting), the lack of a true validation partition introduces a risk of subtle "test leakage" and potential overestimation of performance.

4. **Limited External Validation**
    - No prospective or retrospective external validation was performed using independent clinical data. Without testing on diverse hospital data, concerns remain about robustness under distributional shifts (different patient demographics, radiographic protocols).
5. **Simplified Clinical Setting**
    - The system focuses exclusively on posterior-anterior frontal chest X-rays. Cases requiring lateral projections or additional clinical context (symptoms, lab tests) are not addressed.
    - The ChatGPT assistant, while useful for generic advice, does not replace expert clinical decision making. Its responses reflect the model's understanding of general medical knowledge and may not align perfectly with local clinical guidelines.

5.4 Future Work

To address the limitations above and expand the system's clinical applicability, we propose the following directions:

1. **Multi-Center Data Collection and Validation**
    - Collaborate with multiple hospitals (nationally or internationally) to collect a more diverse dataset covering different X-ray machines, patient demographics, and imaging protocols.
    - Conduct a formal external validation study to quantify model performance across varied populations, measuring sensitivity, specificity, positive predictive value, and negative predictive value in real-world settings.
2. **Expand to Additional Lung Disease Categories**
    - Extend the classification schema to include other pulmonary conditions such as Tuberculosis, COVID-19, Pneumothorax, Pleural Effusion, or Interstitial Lung Disease. This will require new data collection and annotation, as well as multi-label training methodologies (e.g., hierarchical classification or multi-task learning).
3. **Integrate Automatic Lung Segmentation**

- Incorporate a lightweight segmentation network (e.g., U-Net) to automatically crop or mask lung fields before classification. Segmentation can reduce spurious activations (machine labels, ribs, non-pulmonary areas) and improve both classification accuracy and Grad-CAM interpretability.

4. **Model Compression and Edge Deployment**
    - Explore model pruning, quantization, or knowledge distillation to reduce Swin model size and inference latency, enabling deployment on resource-constrained devices (e.g., mobile phones, embedded systems).
    - Convert the trained model to ONNX or TorchScript format for faster, hardware-accelerated inference on Intel/ARM CPUs or NPUs.
5. **Establish a Robust Validation Pipeline**
    - Reserve 10 %–15 % of the training data as a "true" validation set for hyperparameter tuning and early stopping. Implement k-fold cross-validation on the remaining 85 % to ensure no test set leakage.
    - Publish detailed validation metrics (confusion matrices and ROC curves for each fold) to demonstrate stability and reproducibility.
6. **Enhance Chatbot Reliability and Contextualization**
    - Fine-tune a domain-specific language model on radiology reports (e.g., public datasets of chest X-ray captions) so the chatbot can offer more precise, evidence-based explanations.
    - Implement a fallback rule-based FAQ engine to handle frequent, standard questions if the OpenAI API is unavailable or rate-limited.
7. **Clinical Workflow Integration**
    - Develop integration modules for popular radiology information systems (RIS) or picture archiving and communication systems (PACS) to facilitate seamless uploading of new X-rays and automatic result retrieval.
    - Implement user authentication, audit logging, and access controls to comply with data-protection regulations (e.g., HIPAA, GDPR).
8. **User Experience and Accessibility Improvements**
    - Enhance the web interface to include interactive Grad-CAM controls: allow users to adjust heatmap opacity, select color palettes, and zoom into regions of interest.

- Add multilingual support (e.g., English, Vietnamese) for all UI text and chatbot responses. Provide a “dark mode” and adhere to WCAG 2.1 guidelines for users with visual impairments.

9. **Continuous Learning and Model Update Mechanisms**

    - Implement a “clinician feedback” loop: allow radiologists to correct AI predictions. Store these corrections to periodically retrain or fine-tune the model, ensuring continuous improvement and adaptation to new imaging patterns.
    - Set up an automated pipeline for versioning model weights and tracking performance drift over time (monitoring for decreased accuracy as imaging protocols evolve).

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APPENDICES

1. **Source Code:** https://github.com/joshchan1301/pulmo-vision.git
2. **Source Code backend:** https://github.com/joshchan1301/pulmo-backend.git
3. **Swin base model:**

https://drive.google.com/file/d/1VD2pXT9aDHmGwn2KiD3aSoXj5nBglVLw/view?usp=sharing